MATH 101

Homework 5

Polynomial, rational, and radical functions

Sections 5.2-5.7 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

Sections 5.2-5.3 - Power and polynomial functions

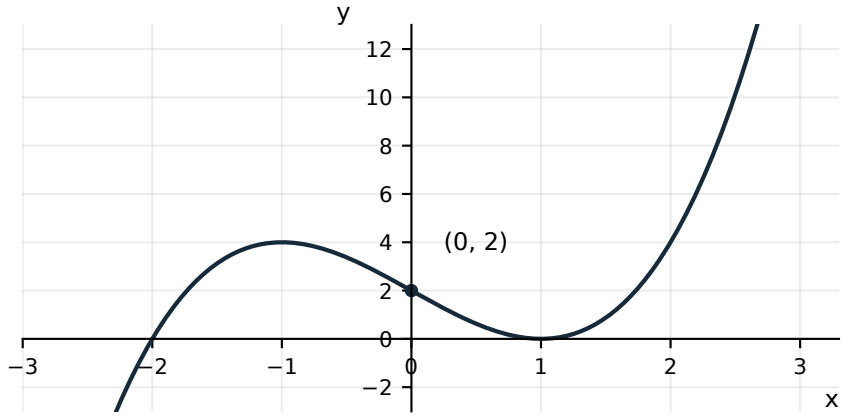
1. Identify which functions are power functions and which are polynomial functions. For each polynomial, give its degree and leading coefficient. Describe the end behavior of each function.

$$f(x) = 3x^4, \qquad g(x) = 3x^4 - 2, \qquad h(x) = \frac{2}{x^2}$$

2. State the degree, leading coefficient, end behavior, zeros, and multiplicities. At each zero, does the graph cross or touch the axis?

$$p(x) = -2(x+1)^2(x-3)^3$$

3. The graph is a cubic. Its only zeros are -2 (crossing) and 1 (touching); the labeled point is on the graph. Write its equation in factored form.



Sections 5.4-5.5 - Polynomial division and zeros

4. List all Rational Root Theorem candidates. Test only -1, 1, and 2. Use the rational zero you find and polynomial long division, then the quadratic formula, to find every zero.

$$p(x) = x^3 - 2x^2 - 2x + 4$$

5. A real-coefficient polynomial has degree 4 and zeros 2 (multiplicity 2) and 1 + i. State its remaining zero and explain why your list is complete.

Section 5.6 - Rational functions

6. Find the domain, hole (if any), vertical and horizontal asymptotes, and intercepts. Use the original restrictions.

$$r(x) = \frac{(x-2)(x+1)}{(x-2)(x-3)}$$

7. Use polynomial long division to find the slant asymptote. Give the domain and vertical asymptote.

$$s(x) = \frac{x^2 + x + 1}{x - 1}$$

8. Use a sign chart to solve. Explain whether each boundary point is included.

$$\frac{x-1}{x+2} \leq 0$$

Section 5.7 - Inverses and radical functions

9. Find the domain, range, both intercepts, and end behavior; sketch the graph.

$$h(x) = 2\sqrt{x+1} - 4$$

10. Find the inverse, with its domain and range.

$$f(x) = (x+1)^2, \qquad x \leq -1$$